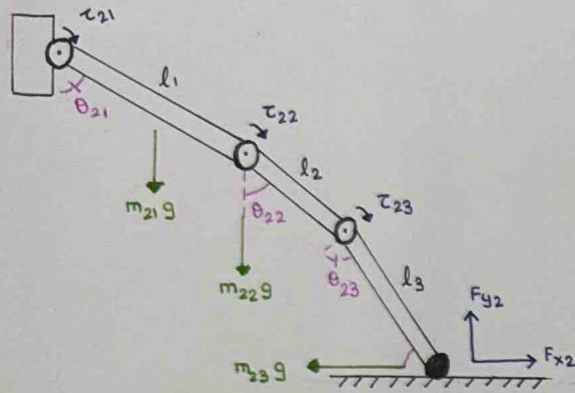


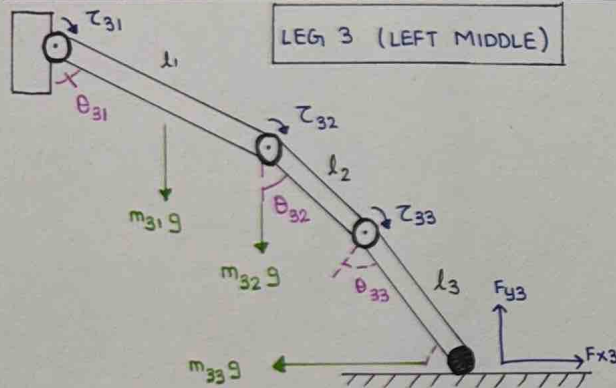
FREE BODY DIAGRAM (FBD) OF ALL SIX LEGS

HEXAPOD ROBOT (3-DOF LEGS: COXA - FEMUR - TIBIA)

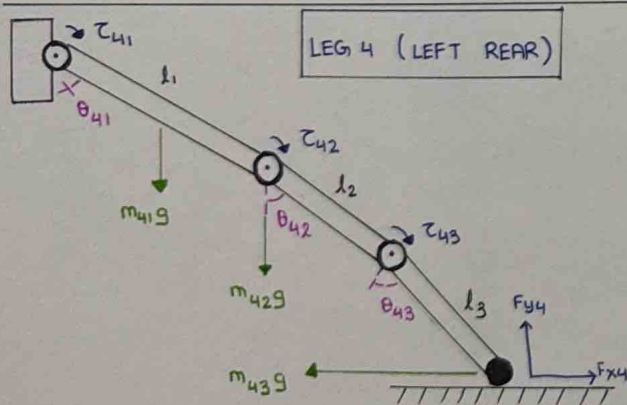
LEG 2 (LEFT FRONT)



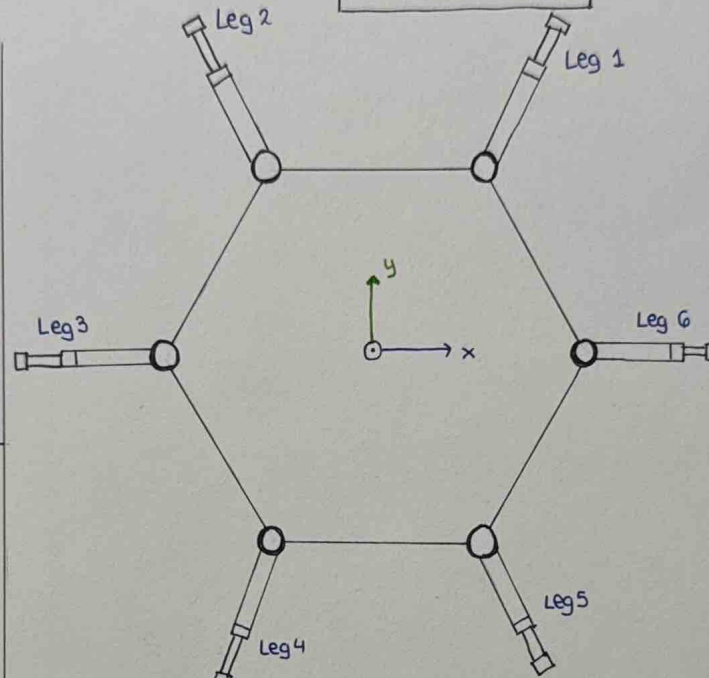
LEG 3 (LEFT MIDDLE)



LEG 4 (LEFT REAR)



TOP / BODY VIEW



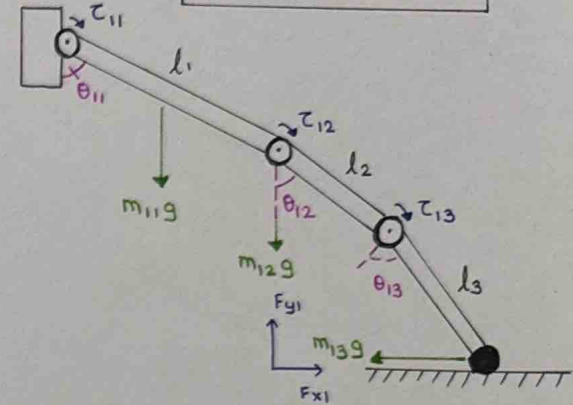
(Numbers show leg location seen from top)

LEGEND:

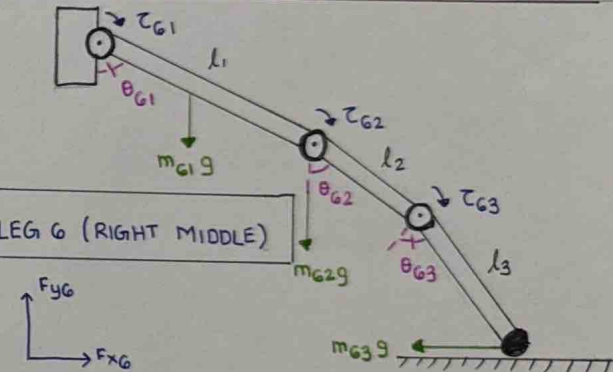
- l_1 = Coxa length
- l_2 = Femur length
- l_3 = Tibia length
- θ_{i1} = Coxa joint angle
- θ_{i2} = Femur joint angle
- θ_{i3} = Tibia joint angle
- ($i = 1$ to 6 for leg no.)
- τ_{i1} = Coxa joint torque
- τ_{i2} = Femur joint torque
- τ_{i3} = Tibia joint torque
- m_{ij} = Mass of link j of leg i
- $m_{ij}g$ = Gravity force on link j of leg i (acting downward)
- F_{xi} = Horizontal ground reaction force (friction)
- F_{yi} = Vertical ground reaction force (normal)

All forces are shown acting on the leg.
(+ve directions: $F_x \rightarrow$ right, $F_y \uparrow$ up)

LEG 1 (RIGHT FRONT)



LEG 6 (RIGHT MIDDLE)



LEG 5 (RIGHT REAR)

